BIN CHEN

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Phone: 480-310-8682 ⋄ Email: bin.chen@northwestern.edu ⋄ Website: binchen.ac

EDUCATION

Arizona State University, Tempe, AZ — 2014 – 2018
Ph.D. Materials Science and Engineering

Fuzhou University, China — 2010 – 2014
B.Eng. Materials Science and Engineering

RESEARCH EXPERIENCE

Northwestern University, Evanston, IL — 2025 – present
Director of Research, Paula M. Trienens Institute for Sustainability and Energy

Northwestern University, Evanston, IL — 2024 – 2025
Associate Director, TEAMUP Consortium
Research: Tandems for Efficient and Advanced Modules using Ultrastable Perovskites

Northwestern University, Evanston, IL — 2022 – present
Research Associate Professor, Department of Chemistry; Department of Electrical & Computer Engineering
Research: Emerging semiconductors for solar and sensing applications

University of Toronto, Canada — 2018 – 2022
Postdoctoral Fellow, Department of Electrical & Computer Engineering
Advisor: Prof. Ted Sargent
Research: Perovskite-based tandem solar cells and quantum dot infrared photodetectors

Arizona State University, Tempe, AZ — 2014 – 2018
Graduate Research Associate, School for Engineering of Matter, Transport & Energy
Advisor: Prof. Sefaattin Tongay
Dissertation: Atomic Scale Characterizations of Two-dimensional Anisotropic Materials and Their Heterostructures
Committee Members: Prof. Sefaattin Tongay, Prof. Mariana Bertoni, Prof. Shery Chang

RESEARCH INTERESTS

Our research group is dedicated to designing innovative energy materials and investigating nano-scale processes within these materials and devices. By integrating principles from physics, chemistry, and materials science, we employ advanced microscopy and spectroscopy techniques for detailed characterization. Our goal is to advance energy technologies and contribute to a sustainable future through cutting-edge research and interdisciplinary collaboration.

SELECTED PUBLICATIONS

† equal contribution, * corresponding

24. Ban, H. W.; Li, X.; Zeiske, S.; Steele, J. A.; López-Arteaga, R. E.; Lu, H.; Zou, Y.; Han, M. G.; Kim, T.-G.; Liu, C.; **Chen, B.***; Sargent, E. H.* Short-Chain Acids Sustain InAs Colloidal Quantum Dot Growth during Synthesis, Extending Spectral Response into the Deep Short-Wave Infrared. *J. Am. Chem. Soc.* **2026**, *148* (21), 21885–21894.

23. Hou, J.; Gilley, I. W.; Wiggins, T. E.; Liu, C.; Yang, Y.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Synthetic Routes to Advancing Perovskite Solar Cells Through Interface Design. *Nat. Synth.* **2026**. https://doi.org/10.1038/s44160-026-01033-4.

22. Zeiske, S.; Ban, H. W.; Li, X.; Deng, B.; López-Arteaga, R.; Kazianga, U. H.; Han, M. G.; Kim, T.-G.; **Chen, B.***; Sargent, E. H.* Revealing Additional Size-Dependent Defect Suppression Channels Governing Detectivity in InAs Colloidal Quantum Dot Photodiodes. *Nano Lett.* **2025**, *25* (48), 16862–16868.

21. Huang, C.; Yang, Y.; Liu, C.; Chen, H.; Chaudhuri, S.; Jeon, W. C.; Li, M.; Rolston, N.; Bati, A. S. R.; Gilley, I. W.; Kumral, B.; Serles, P.; Filleter, T.; Schatz, G. C.; Kanatzidis, M. G.; **Chen, B.***; Chen, L. X.*; Sargent, E. H.* Electrostatically Enhanced Buried Interface Binding of Self-Assembled Monolayers for Efficient and Stable Inverted Perovskite Solar Cells. *Adv. Mater.* **2025**, *37* (43), e08740.

20. Bati, A. S. R.; Liu, C.; Gilley, I. W.; Musgrave, C. B., III; Maxwell, A.; Steele, J. A.; Yang, Y.; Chen, H.; Wan, H.; Xu, J.; Solano, E.; Zhang, R.; Huang, C.; Rehl, B.; Lempesis, N.; Carnevali, V.; Vezzosi, A.; Zeng, L.; Grater, L.; Li, M.; Rolston, N.; Choi, D.; Sláma, V.; Rothlisberger, U.; Wang, L.; Goddard, W. A., III; Kanatzidis, M. G.; **Chen, B.***; Bakr, O. M.*; Sargent, E. H.* A Chemically Bonded Monolayer Interface Enables Enhanced Thermal Stability and Efficiency in Pb-Sn Perovskite Solar Cells. *Joule* **2025**, *9* (9), 102047.

19. Choi, D.; Shin, D.; Li, C.; Liu, Y.; Bati, A. S. R.; Kachman, D. E.; Yang, Y.; Li, J.; Lee, Y. J.; Li, M.; Penukula, S.; Kim, D. B.; Shin, H.; Chen, C.-H.; Park, S. M.; Liu, C.; Maxwell, A.; Wan, H.; Rolston, N.; Sargent, E. H.*; **Chen, B.*** Carboxyl-Functionalized Perovskite Enables ALD Growth of a Compact and Uniform Ion Migration Barrier. *Joule* **2025**, 9 (3), 101801.

18. Yang, Y.; Chen, H.; Liu, C.; Xu, J.; Huang, C.; Malliakas, C. D.; Wan, H.; Bati, A. S. R.; Wang, Z.; Reynolds, R. P.; Gilley, I. W.; Kitade, S.; Wiggins, T. E.; Zeiske, S.; Suragtkhuu, S.; Batmunkh, M.; Chen, L. X.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Amidination of Ligands for Chemical and Field-Effect Passivation Stabilizes Perovskite Solar Cells. *Science* 2024, *386* (6724), 898–902.

17. Li, C.†; Chen, L.†; Jiang, F.†; Song, Z.†; Wang, X.; Balvanz, A.; Ugur, E.; Liu, Y.; Liu, C.; Maxwell, A.; Chen, H.; Liu, Y.; Wang, Z.; Xia, P.; Li, Y.; Fu, S.; Sun, N.; Grice, C. R.; Wu, X.; Fink, Z.; Hu, Q.; Zeng, L.; Jung, E.; Wang, J.; Park, S. M.; Luo, D.; Chen, C.; Shen, J.; Han, Y.; Perini, C. A. R.; Correa-Baena, J.-P.; Lu, Z.-H.; Russell, T. P.; De Wolf, S.; Kanatzidis, M. G.; Ginger, D. S.; **Chen, B.***; Yan, Y.*; Sargent, E. H.* Diamine Chelates for Increased Stability in Mixed Sn–Pb and All-Perovskite Tandem Solar Cells. *Nat. Energy* 2024, *9* (11), 1388–1396.

16. Chen, H.†; Liu, C.†; Xu, J.†; Maxwell, A.†; Zhou, W.†; Yang, Y.; Zhou, Q.; Bati, A. S. R.; Wan, H.; Wang, Z.; Zeng, L.; Wang, J.; Serles, P.; Liu, Y.; Teale, S.; Liu, Y.; Saidaminov, M.; Hoogland, S.; Filleter, T.; Kanatzidis, M. G.; **Chen, B.***; Ning Z*; Sargent, E. H.* Improved charge extraction in inverted perovskite solar cells with dual-site-binding ligands. *Science* 2024, 384 (6692), 189–193.

15. Xu, J.; Maxwell, A.; Song, Z.; Bati, A. S. R.; Chen, H.; Li, C.; Park, S. M.; Yan, Y.; **Chen, B.***; Sargent, E. H.* The Dynamic Adsorption Affinity of Ligands Is a Surrogate for the Passivation of Surface Defects. *Nat. Commun.* 2024, 15 (1), 2035.

14. Maxwell, A.†; Chen, H.†; Grater, L.; Li, C.; Teale, S.; Wang, J.; Zeng, L.; Wang, Z.; Park, S. M.; Vafaie, M.; Sidhik, S.; Metcalf, I. W.; Liu, Y.; Mohite, A. D.; **Chen, B.***; Sargent, E. H.* All-Perovskite Tandems Enabled by Surface Anchoring of Long-Chain Amphiphilic Ligands. *ACS Energy Lett.* 2024, 520–527.

13. Yang, Y.†; Liu, C.†; Ding, Y.†; Ding, B.†; Xu, J.†; Liu, A.; Yu, J.; Grater, L.; Zhu, H.; Hadke, S. S.; Sangwan, V. K.; Bati, A. S. R.; Hu, X.; Li, J.; Park, S. M.; Hersam, M. C.; **Chen, B.***;

Nazeeruddin, M. K.*; Kanatzidis, M. G.*; Sargent, E. H.* A Thermotropic Liquid Crystal Enables Efficient and Stable Perovskite Solar Modules. *Nat. Energy* 2024, 1–8.

12. Liu, C.†; Yang, Y.†; Chen, H.†; Xu, J.†; Liu, A.†; Bati, A. S. R.; Zhu, H.; Grater, L.; Hadke, S. S.; Huang, C.; Sangwan, V. K.; Cai, T.; Shin, D.; Chen, L. X.; Hersam, M. C.; Mirkin, C. A.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Bimolecularly-passivated interface enables efficient and stable inverted perovskite solar cells, *Science* 2023, 382 (6672), 810–815.

11. Liu, Y.†; Zhu, T.†; Grater, L.†; Chen, H.†; Reis, R.; Maxwell, A.; Cheng, M.; Dong, Y.; Teale, S.; Leontowich, A. F. G.; Kim, C.; Chan, P. T.; Wang, M.; Paritmongkol, W.; Gao, Y.; Park, S.; Xu, J.; Khan, J. I.; Laquai, F.; Walker, G. C.; Dravid, V. P.; **Chen, B.***; Sargent, E. H.* A Three-Dimensional Quantum Dot Network Stabilizes Perovskite Solids via Hydrostatic Strain. *Matter* 2024, 7 (1), 107–122.

10. Chen, H.†; Maxwell, A.†; Li, C.†; Teale, S.†; **Chen, B.†**; Zhu, T.; Ugur, E.; Harrison, G.; Grater, L.; Wang, J.; Wang, Z.; Zeng, L.; Park, S. M.; Chen, L.; Serles, P.; Awni, R. A.; Subedi, B.; Zheng, X.; Xiao, C.; Podraza, N. J.; Filleter, T.; Liu, C.; Yang, Y.; Luther, J. M.; De Wolf, S.; Kanatzidis, M. G.; Yan, Y.; Sargent, E. H. Regulating Surface Potential Maximizes Voltage in All-Perovskite Tandems. *Nature* 2023, 613 (7945), 676–681.

9. Chen, H.†; Teale, S.†; **Chen, B.†**; Hou, Y.†; Grater, L.; Zhu, T.; Bertens, K.; Park, S. M.; Atapattu, H. R.; Gao, Y.; Wei, M.; Johnston, A. K.; Zhou, Q.; Xu, K.; Yu, D.; Han, C.; Cui, T.; Jung, E. H.; Zhou, C.; Zhou, W.; Proppe, A. H.; Hoogland, S.; Laquai, F.; Filleter, T.; Graham, K. R.; Ning, Z.; Sargent, E. H. Quantum-Size-Tuned Heterostructures Enable Efficient and Stable Inverted Perovskite Solar Cells. *Nat. Photonics* 2022, 16 (5), 352–358.

8. **Chen, B.**; Sargent, E. H. What Does Net Zero by 2050 Mean to the Solar Energy Materials Researcher? *Matter* 2022, 5 (5), 1322–1325.

7. **Chen, B.**; Chen, H.; Hou, Y.; Xu, J.; Teale, S.; Bertens, K.; Chen, H.; Proppe, A.; Zhou, Q.; Yu, D.; Xu, K.; Vafaie, M.; Liu, Y.; Dong, Y.; Jung, E. H.; Zheng, C.; Zhu, T.; Ning, Z.; Sargent, E. H. Passivation of the Buried Interface via Preferential Crystallization of 2D Perovskite on Metal Oxide Transport Layers. *Adv. Mater.* 2021, e2103394.

6. Fang, Z.†; Wang, L.†; Mu, X.†; **Chen, B.†**; Xiong, Q.; Wang, W. D.; Ding, J.; Gao, P.; Wu, Y.; Cao, J. Grain Boundary Engineering with Self-Assembled Porphyrin Supramolecules for Highly Efficient Large-Area Perovskite Photovoltaics. *J. Am. Chem. Soc.* 2021.

5. Jung, E. H.†; **Chen, B.†**; Bertens, K.; Vafaie, M.; Teale, S.; Proppe, A.; Hou, Y.; Zhu, T.; Zheng, C.; Sargent, E. H. Bifunctional Surface Engineering on SnO2 Reduces Energy Loss in Perovskite Solar Cells. *ACS Energy Lett.* 2020, 5 (9), 2796–2801.

4. **Chen, B.**; Baek, S.-W.; Hou, Y.; Aydin, E.; De Bastiani, M.; Scheffel, B.; Proppe, A.; Huang, Z.; Wei, M.; Wang, Y.-K.; Jung, E.-H.; Allen, T. G.; Van Kerschaver, E.; García de Arquer, F. P.; Saidaminov, M. I.; Hoogland, S.; De Wolf, S.; Sargent, E. H. Enhanced Optical Path and Electron Diffusion Length Enable High-Efficiency Perovskite Tandems. *Nat. Commun.* 2020, 11 (1), 1257.

3. Manekkathodi, A.†; **Chen, B.†**; Kim, J.; Baek, S.-W.; Scheffel, B.; Hou, Y.; Ouellette, O.; Saidaminov, M. I.; Voznyy, O.; Madhavan, V. E. Solution-Processed Perovskite-Colloidal Quantum Dot Tandem Solar Cells for Photon Collection beyond 1000 Nm. *Journal of Materials Chemistry A* 2019, 7 (45), 26020–26028.

2. **Chen, B.**; Wu, K.; Suslu, A.; Yang, S.; Cai, H.; Yano, A.; Soignard, E.; Aoki, T.; March, K.; Shen, Y. Controlling Structural Anisotropy of Anisotropic 2D Layers in Pseudo-1D/2D Material Heterojunctions. *Adv. Mater.* 2017, 29 (34), 1701201.

1. **Chen, B.**; Sahin, H.; Suslu, A.; Ding, L.; Bertoni, M. I.; Peeters, F.; Tongay, S. Environmental Changes in MoTe2 Excitonic Dynamics by Defects-Activated Molecular Interaction. *ACS nano*

2015, 9 (5), 5326–5332.

FUNDING SUPPORT

Pending Research

3. DE-FOA-0003337
   U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy
   $1,500,000
   06/2025 – 05/2028
   Efficient and stable near-infrared organic photovoltaics for hybrid organic tandem solar cells
   Role: Co-Principal Investigator

2. DOI-BAA-SOLSTICE-FY25-01
   IARPA
   $2,00,000
   6/2026 – 12/2027
   Superior Options for Long-life Solar Technologies with Impressive Conversion Efficiencies (SOLSTICE)
   Role: Co-Principal Investigator

1. NSF 23-612
   U.S. National Science Foundation
   $600,000
   05/2025 – 04/2028
   Synthesis and surface chemistry of InAs colloidal quantum dots toward efficient short-wave infrared detection
   Role: Co-Principal Investigator

Current and Completed Research

8. Translation and Incubation Fund
   Trienens Institute for Sustainability and Energy
   $100,000
   01/2024 – 12/2024
   Stable perovskite solar cells with cost-effective bilayer metal oxides as electron transport layers
   Role: Principal Investigator

7. DE-FOA-0003058
   U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy
   $6,700,000
   09/2024 – 08/2026
   STACKED: Stability and Characterization of Hole-Transporting Layers Key to Enabling Outdoor Durability
   Role: Co-Principal Investigator

6. Seed Funding Initiative
   Center for Engineering Sustainability and Resilience
   $80,000
   02/2024 – 8/2025
   Introducing AC Photo-Hall Method: Separating Electron/Hole Mobilities in Perovskite Photovoltaics
   Role: Co-Principal Investigator

5. OSR-5624
   King Abdullah University of Science and Technology
   $350,000

02/2024 – 01/2026
All-Perovskite Tandem Solar Cells
Role: Co-Principal Investigator

4. DE-EE0010502
U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy
$9,000,000
09/2023 – 08/2026
TEAMUP: Tandems for Efficient and Advanced Modules using Ultrastable Perovskites
Role: Senior Personnel

3. HR001122S0044-SNAP-FP-009
U.S. Department of Defense, Defense Advanced Research Projects Agency
$10,000,000
08/2023 – 07/2027
SYNCED: Interfacing Synthetic Biology with Electrochemical Detectors for Smart Non-Invasive Assays of Physiology
Role: Senior Personnel

2. N00014-20-1-2572
U.S. Department of the Navy, Office of Naval Research
$480,000
08/2020 – 07/2023
Wide-bandgap perovskites for efficient, stable tandems
Role: Senior Personnel

1. OSR-2020-CRG9-4350.2
King Abdullah University of Science and Technology, Office of Sponsored Research
$600,000
04/2021 – 03/2024
SOLSTICE: Solar-driven Circular Carbon Enabled by Perovskite/Perovskite/Si Triple-Junction Tandems
Role: Senior Personnel

PATENTS

3. US Patent Application: Passivating Perovskite Optoelectronic Devices (Application No. 19/375,634, 2026)
2. US and Canadian Patent: A Surface Treatment Method to Passivate Inverted Structure Perovskite Solar Cells
1. US Provisional Patent: Perovskite Solar Cells With Dual Site Binding Ligands

INVITED TALKS

21. MRS Fall Meeting, December 2026
20. IPS-25, the 25th International Conference on Photochemical Conversion and Storage of Solar Energy, Seoul, July 2026
19. MRS Spring Meeting, April 2026
18. nanoGe – MATSUS Fall 2025, October 2025
17. Lawrence Symposium on Epitaxy, Arizona State University, October 2025
16. The University of Toledo, October 2025
15. MRS Spring Meeting, April 2025

14. Ludwig Maximilian University of Munich, March 2025
13. Nanyang Technological University, March 2025
12. National University of Singapore, February 2025
11. 4th tandemPV Workshop, June 2024
10. Physics Seminar Series, UC Merced, October 2023
9. 2nd Northwestern/Münster Symposium on Smart Materials, August 2023
8. Homeland Defense & Security Information Analysis Center, April 2023
7. APS March Meeting, March 2023
6. EcoMat Webinar: Perovskite Materials for Photovoltaic and Optoelectronic Applications (virtual), January 2023
5. Lawrence Symposium on Epitaxy, Arizona State University, January 2023
4. Zhejiang University, June 2022
3. KAUST Research Conference: 2022 Accelerating Solar Energy Research towards meeting Vision 2030 Goals, May 2022
2. ICFO – UofT – Stanford International School on the Frontiers of Light, October 2021
1. MRS Spring Meeting (virtual), April 2021

CONFERENCE ORGANIZATION & SERVICE

4. Organizer, ACS Spring Meeting Symposium, March 2026
3. Co-organizer, MRS Spring Meeting Symposium, April 2025
2. Session chair, MRS Fall Meeting, November 2023
1. Co-organizer, ACS Fall Meeting Symposium: Organic, Perovskite and Hybrid Solar (raised $5,000 sponsorship for the symposium), August 2023

SELECTED ORAL & POSTER PRESENTATIONS

6. Oral, MRS Fall Meeting, December 2024
5. Oral, ACS Fall Meeting, August 2024
4. Oral, MRS Fall Meeting, November 2023
3. Oral, IEEE PVSC 50th, June 2023
2. Poster, MRS Spring Meeting, March 2017
1. Poster, MRS Spring Meeting, March 2016

TEACHING AND MENTORING

Instructor at Kellogg and the Querrey InQbation Lab 2024

I teach the one-quarter Independent Study course, the Kellogg – Q Lab Entrepreneurial Residency. This course allows MBA students to be embedded in Northwestern research centers, providing structured entry points into cutting-edge technology areas and offering exposure to cross-disciplinary research.

Instructor for Independent Study (399) at Weinberg College of Arts and Sciences 2024

I teach the on-quarter Independent Study course on optoelectronics, where I am responsible for developing the syllabus and providing guidance to the enrolled undergraduate students. This course offers students opportunities to build fundamental knowledge and explore research trends in the literature.

Perovskite Photovoltaic Research Group Leader 2019 – 2022

I manage a team of over 15 members focusing on perovskite photovoltaics research. My responsibilities include mentoring on specific research projects, conceptualizing manuscripts, writing grant proposals, and preparing grant reports.

Lecturer (simulated) 2019

At the Teaching in Higher Education course at the University of Toronto, I developed my syllabus on "Two-dimensional semiconductor materials and systems" and taught in simulated classes.

ACADEMIC AND SOCIAL SERVICE

Journal advisory board 2025 – present

EES Solar

Journal reviewer 2016 – present

Science, Nature, Nature Energy, Nature Photonics, Nature Sustainability, Chemical Reviews, Journal of the American Chemical Society, Joule, Angewandte Chemie; Nature Communications, Advanced Materials, Matter, Energy & Environmental Science, Advanced Energy Materials, ACS Energy Letters, ACS Nano, ACS Photonics, Advanced Science, Chemical Science, EES Solar, ACS Applied Materials & Interfaces, ACS Applied Energy Materials; The Journal of Physical Chemistry, The Journal of Physical Chemistry Letters, Journal of Applied Physics, Progress in photovoltaics, Journal of Photovoltaics, Journal of Physics D: Applied Physics, Journal of Physics: Condensed Matter, 2D Materials, Nanotechnology, Nano Energy

Proposal reviewer 2023 – present

U.S. National Science Foundation
Natural Sciences and Engineering Research Council of Canada
ACS Petroleum Research Fund
National Research Foundation of Singapore

HONORS AND AWARDS

4. Stanford/Elsevier's list of top 2% scientists worldwide
3. Highly Cited Researcher in the field of Cross-Field - 2023 (Clarivate)
2. Highly Cited Researcher in the field of Cross-Field - 2024 (Clarivate)
1. Highly Cited Researcher in the field of Cross-Field - 2025 (Clarivate)

ALL PUBLICATIONS

142. Ban, H. W.; Li, X.; Zeiske, S.; Steele, J. A.; López-Arteaga, R. E.; Lu, H.; Zou, Y.; Han, M. G.; Kim, T.-G.; Liu, C.; **Chen, B.***; Sargent, E. H.* Short-Chain Acids Sustain InAs Colloidal Quantum Dot Growth during Synthesis, Extending Spectral Response into the Deep Short-Wave Infrared. *J. Am. Chem. Soc.* **2026**, *148* (21), 21885–21894.

141. Artuk, K.; Türkay, D.; Kuba, A.; Riemelmoser, S.; Steele, J. A.; Hurni, J.; Spitznagel, J.; Quest, H.; De Bastiani, M.; Zhao, J.; Diekmann, J.; Ongaro, C.; Othman, M.; Heydarian, M.; Fischer, O.; Lai, H.; Austin, J. S.; Zeiske, S.; López-Arteaga, R.; Liu, C.; Mensi, M. D.; Castro-Mendez, A.-F.; Li, M.; Gries, T. W.; Hill, S.; Saenz, F.; Champault, L.; Can, H. A.; Golobostanfard, M. R.; Desai, U.; Remondeau, P.; Solano, E.; Portale, G.; Faes, A.; Lang, F.; Musiienko, A.; Rolston, N.; Fu, F.; Schubert, M. C.; Schindler, F.; **Chen, B.**; Pasquarello, A.; Sargent, E. H.; Hessler-Wyser, A.; Jeangros, Q.; Ballif, C.; Wolff, C. M. Triple-Junction Solar Cells with Improved Carrier and Photon Management. *Nature* **2026**. https://doi.org/10.1038/s41586-026-10385-y.

140. Hou, J.; Fletcher, J.; Hall, S. J.; Zhang, H.; Zacharias, M.; Volonakis, G.; Welton, C.; Zeiske, S.; Gilley, I. W.; Shin, D.; Mandani, F.; Metcalf, I.; Sun, S.; Zhang, B.; Guo, Y.; **Chen, B.**;

Reddy, G. N. M.; Katan, C.; Even, J.; Sfeir, M. Y.; Kanatzidis, M. G.; Mohite, A. D. Exciton Diffusion beyond 2 m Enabled by Maximum Symmetry in Two-Dimensional Perovskites. *Nat. Synth.* **2026**. https://doi.org/10.1038/s44160-026-01041-4.

139. Choi, D.; Kim, J.; Jaffer, S.; **Chen, B.**; Xie, K.; Sargent, E. H. Translating Insights from Progress in Photovoltaics to Accelerate Industrial-Scale CO2 Electroreduction. *Nat. Energy* **2026**. https://doi.org/10.1038/s41560-025-01953-z.

138. Hou, J.; Gilley, I. W.; Wiggins, T. E.; Liu, C.; Yang, Y.; **Chen, B.***; Kanatzidis, M. G.*; Sargent, E. H.* Synthetic Routes to Advancing Perovskite Solar Cells Through Interface Design. *Nat. Synth.* **2026**. https://doi.org/10.1038/s44160-026-01033-4.

137. Zeiske, S.; Ban, H. W.; Li, X.; Deng, B.; López-Arteaga, R.; Kazianga, U. H.; Han, M. G.; Kim, T.-G.; **Chen, B.***; Sargent, E. H.* Revealing Additional Size-Dependent Defect Suppression Channels Governing Detectivity in InAs Colloidal Quantum Dot Photodiodes. *Nano Lett.* **2025**, *25* (48), 16862–16868.

136. Huang, C.; Yang, Y.; Liu, C.; Chen, H.; Chaudhuri, S.; Jeon, W. C.; Li, M.; Rolston, N.; Bati, A. S. R.; Gilley, I. W.; Kumral, B.; Serles, P.; Filleter, T.; Schatz, G. C.; Kanatzidis, M. G.; **Chen, B.***; Chen, L. X.*; Sargent, E. H.* Electrostatically Enhanced Buried Interface Binding of Self-Assembled Monolayers for Efficient and Stable Inverted Perovskite Solar Cells. *Adv. Mater.* **2025**, *37* (43), e08740.

135. Bati, A. S. R.; Liu, C.; Gilley, I. W.; Musgrave, C. B., III; Maxwell, A.; Steele, J. A.; Yang, Y.; Chen, H.; Wan, H.; Xu, J.; Solano, E.; Zhang, R.; Huang, C.; Rehl, B.; Lempesis, N.; Carnevali, V.; Vezzosi, A.; Zeng, L.; Grater, L.; Li, M.; Rolston, N.; Choi, D.; Sláma, V.; Rothlisberger, U.; Wang, L.; Goddard, W. A., III; Kanatzidis, M. G.; **Chen, B.***; Bakr, O. M.*; Sargent, E. H.* A Chemically Bonded Monolayer Interface Enables Enhanced Thermal Stability and Efficiency in Pb-Sn Perovskite Solar Cells. *Joule* **2025**, *9* (9), 102047.

134. Hidayatova, L.; Mi, C.; Akhmedov, N. G.; Liu, Y.; Shafiq, A. K.; Afshari, H.; Mohamed-Raseek, N.; Popy, D. A.; Xiang, S.; Chen, Y.-C.; Saparov, B.; Peters, J. W.; Talapin, D. V.; **Chen, B.**; Furis, M.; Glaser, E. R.; Dong, Y. Efficient Mn^{2+} Doping in Non-Stoichiometric Cesium Lead Bromide Perovskite Quantum Dots. *J. Am. Chem. Soc.* **2025**, *147* (38), 35069–35080.

133. Fu, S.; Sun, N.; Chen, H.; Liu, C.; Wang, X.; Li, Y.; Abudulimu, A.; Xu, Y.; Ramakrishnan, S.; Li, C.; Yang, Y.; Wan, H.; Huang, Z.; Xian, Y.; Yin, Y.; Zhu, T.; Chen, H.; Rahimi, A.; Saeed, M. M.; Zhang, Y.; Yu, Q.; Ginger, D. S.; Ellingson, R. J.; **Chen, B.**; Song, Z.; Kanatzidis, M. G.; Sargent, E. H.; Yan, Y. On-Demand Formation of Lewis Bases for Efficient and Stable Perovskite Solar Cells. *Nat. Nanotechnol.* **2025**, *20* (6), 772–778.

132. Choi, D.; Shin, D.; Li, C.; Liu, Y.; Bati, A. S. R.; Kachman, D. E.; Yang, Y.; Li, J.; Lee, Y. J.; Li, M.; Penukula, S.; Kim, D. B.; Shin, H.; Chen, C.-H.; Park, S. M.; Liu, C.; Maxwell, A.; Wan, H.; Rolston, N.; Sargent, E. H.*; **Chen, B.*** Carboxyl-Functionalized Perovskite Enables ALD Growth of a Compact and Uniform Ion Migration Barrier. *Joule* **2025**, *9* (3), 101801.

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